

Formal Feature Attribution as a Causal Calculator for Drug-Drug Interaction Risk in Polypharmacy: A Systems Pharmacology Approach Using Virginia's All Payer Claim Database

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Abstract

Background: Polypharmacy (≥ 5 concurrent medications) in older adults (≥ 65 years) drives adverse drug events (ADEs) and emergency department (ED) utilization, yet existing DDI databases characterize pairwise interactions in isolation, failing to capture synergistic multi-drug effects. **Methods:** Using a state-level All-Payer Claims Database (APCD; 6,929,576 unique patients; 2016–2019), we constructed a non-opioid ED cohort spanning all age bands (0–114 years; $n = 54,247$ total cases across 8 bands; 5:1 matched controls), with primary causal analysis focused on geriatric bands (65–114 years; $n = 1,182$ cases, 16,105 controls). A 30-day causality window isolated proximal drug exposures preceding each ED visit. A per-event-density-bin ensemble (CatBoost, XGBoost, XGBoost-RF; Optuna-tuned) was trained on 2016–2018 data (holdout: 2019). Raw event totals define density stratum and routing for those models; the learned inputs use ordinal density encoding, not continuous counts (Chapter~2). Formal Feature Attribution (FFA) was extended to multi-feature interaction testing (pairs and triplets), evaluating synergy via counterfactual probability shifts and Intervention Rate (IR) scores for deprescribing prioritization. **Results:** All 8 age bands achieved PR-AUC ≥ 0.908 (range: 0.908–0.997; monotonically increasing with age, peaking at 0.997 in the 75–84 band), contrasting with the inverted-U pattern of the opioid ED cohort. FFA identified 115 synergistic drug pairs and 5,021 high-risk triplets. Top pairs: acetaminophen + levofloxacin (IE = 16.3; CYP1A2-mediated) and levofloxacin + lorazepam (IE = 11.9; QT + CNS depression) in the 85–114 band. Top IR targets: simvastatin (IR 7.0×10^{-4} ; E78.5 hyperlipidemia; Beers/CYP3A4), furosemide (IR 2.0×10^{-4} ; I50.9 heart failure), and alprazolam (IR 1.0×10^{-4} ; F41.1 anxiety; Beers CNS). Z-code analysis revealed a U-shaped monitoring-risk relationship: moderate preventive monitoring (Z71.89, Z00.00; 1–12% of claims; Q2) was protective (crude OR = 0.25; 95% CI 0.18–0.34; $P < 0.001$), while patients with extreme monitoring proportions (Q4, $\geq 12\%$) had equivalent risk to unmonitored patients. **Conclusions:** FFA multi-feature interaction analysis constitutes a “causal calculator” for polypharmacy risk, detecting synergistic drug combinations invisible to pairwise DDI databases and quantifying deprescribing priorities as actionable, cohort-specific IR scores across the full age spectrum from pediatric to geriatric ED presentations.

KEYWORDS

polypharmacy, drug-drug interactions, Formal Feature Attribution, causal inference, systems pharmacology, geriatrics

1 | INTRODUCTION

Polypharmacy (concurrent use of ≥ 5 medications¹) affects more than 40% of adults aged ≥ 65 years in the United States²,

with prevalence rising to over 67% in adults aged ≥ 75 years. In older adults, polypharmacy drives adverse drug events (ADEs), accounting for an estimated 100,000 hospitalizations and 28,000 deaths annually³. A prospective UK cohort study found that 6.5% of medical admissions were directly attributable to ADEs, with DDIs responsible for 11% of those admissions and polypharmacy burden (drug count) the strongest independent predictor⁴. Polypharmacy-related ED utilization is partially

Abbreviations: ADE, adverse drug event; APCD, All-Payer Claims Database; AUC-PR, area under the precision-recall curve; AUROC, area under the receiver operating characteristic curve; DDI, drug-drug interaction; ED, emergency department; FFA, Formal Feature Attribution; HCG, Healthcare Cost Group; ICD, International Classification of Diseases; IR, Intervention Rate; MCCV, Monte Carlo cross-validation; OR, odds ratio; PGx, pharmacogenomics; SHAP, SHapley Additive exPlanations

preventable through proactive medication review, but the clinical tools available for polypharmacy management have not kept pace with the complexity of modern multi-drug regimens in geriatric patients.

The pharmacogenomic dimension of polypharmacy risk is systematically understudied in geriatric populations. Older adults experience age-related declines in CYP enzyme activity (particularly CYP3A4 and CYP2C9), renal clearance, and hepatic blood flow that alter drug metabolism in ways that existing DDI databases do not fully capture⁵. A 12-gene pharmacogenetic panel study demonstrated that PGx-guided prescribing reduced actionable ADEs by 30% in a European RCT⁶, confirming that genomically-informed medication review has measurable clinical impact even in real-world settings. Yet no published claims-based study has combined PGx interaction scoring with multi-feature DDI detection in a geriatric polypharmacy cohort.

Current approaches to polypharmacy management rely on pairwise DDI databases (e.g., Drugs.com, Micromedex) that flag co-prescribed drug pairs based on known pharmacokinetic or pharmacodynamic mechanisms. These databases are useful but limited in a specific way:

1. They characterize *pairwise* interactions, ignoring synergistic multi-drug effects (e.g., triple combinations where each pairwise interaction appears benign).
2. They do not prioritize interventions by expected clinical impact; every flagged pair receives the same alert regardless of the expected probability of a serious adverse event.
3. They cannot account for patient-level comorbidity context or the pharmacokinetic consequences of combined renal and hepatic impairment in the oldest-old.

A systems pharmacology perspective demands a “causal calculator” that simultaneously: (a) identifies multi-drug combinations with *synergistic* ADE risk exceeding additive pairwise predictions; (b) quantifies each drug’s causal contribution to population-level risk; and (c) ranks deprescribing priorities by expected risk reduction, enabling pharmacists to target the highest-yield intervention from a finite clinical review budget.

This study extends Formal Feature Attribution (FFA) from the Chapter 2 Consensus Filter to multi-feature interaction testing, evaluating feature pairs and triplets for synergistic and antagonistic effects, applied to a geriatric non-opioid ED cohort derived from the Virginia APCD. The 30-day causality window isolates proximal drug exposures responsible for acute ADE events, distinguishing this approach from annual medication reconciliation tools that cannot resolve acute DDI timing.

1.1 | Study Objectives

1. Apply FFA multi-feature interaction testing to identify synergistic DDI combinations in the 65–114 year polypharmacy cohort.
2. Quantify Intervention Rate (IR) scores as deprescribing priorities for the highest-risk drug-drug combinations.
3. Characterize the “hairball” of drug co-occurrence networks via FP-Growth visualization to contextualize FFA causal findings.
4. Investigate the relationship between high transaction density (Z-codes) and ADE risk to distinguish managed from unmanaged polypharmacy.

2 | METHODS

Utilization density (read first). The ensemble risk models use the same convention as Chapter~2 (*Ensemble Modeling*: utilization density): total pre-index events per patient determine stratum assignment and which per-stratum model is trained or loaded; the **gradient-boosting** matrix includes `n_event_bin_ordinal` (0–3) **without** the raw `n_events` count. **This chapter’s** Z-code proportion and logistic-regression analyses (§Results) use **raw** 30-day claim counts as statistical covariates where stated; that is a **different** statistical use of counts than the GBM feature matrix.

2.1 | Study Design and Data Source

This was a retrospective matched case-control study using Virginia’s All-Payer Claims Database (APCD) spanning calendar years 2016–2019, obtained under a data-use agreement (DUA) with the Virginia Center for Health Innovation (VCHI) and Virginia Health Information (VHI). Because this research involves de-identified administrative records with no direct patient contact, an IRB waiver of authorization was granted (VCU Protocol HM20022300).

The study population was restricted to adults aged 65 years and older, representing the geriatric polypharmacy cohort defined in Chapter 2. The analysis window was the 30-day period immediately preceding each patient’s index ED visit (for cases) or matched index date (for controls), a window selected to capture proximal drug exposures responsible for acute adverse drug events rather than chronic long-term prescription patterns. All features were derived from 2016–2018 claims only; the 2019 claims year served as the strictly withheld prospective temporal holdout for model evaluation, consistent with the temporal validation strategy applied across all dissertation chapters.

2.2 | Cohort Construction

Case definition: Adults aged 65–114 years with at least one emergency department visit identified by Milliman Healthcare Cost Group (HCG) codes O11 (Emergency Room) or P51 (Emergency Room Visit) on medical claims with an index date between January 1, 2016 and December 31, 2019. All patients with any opioid-related ICD-10 codes (F11.xx, T40.xx) in any diagnosis field across all claims in the analysis period were strictly excluded; this exclusion ensures the study measures polypharmacy-related adverse drug events attributable to drug-drug interactions, not opioid use disorder co-morbidity.

Patients were required to have at least one pharmacy claim in the 30-day lookback window (minimum drug exposure for meaningful DDI testing) and continuous APCD enrollment for 30 days prior to the index date.

Control definition: Adults aged 65–114 years with no ED visits (no HCG O11/P51 claims in the analysis period) and no opioid-related codes, matched 5:1 to cases without replacement using nearest-neighbor matching within age band and sex. Controls were randomly assigned an index date sampled from within their matching calendar year to anchor the 30-day lookback window.

Age bands: Three geriatric age bands were analyzed separately: 65–74 years (young-old), 75–84 years (middle-old), and 85–114 years (oldest-old), reflecting clinically distinct pharmacokinetic profiles (declining renal clearance, body composition changes, and polypharmacy burden escalation across the geriatric age spectrum).

2.3 | Feature Engineering

Primary feature domain: Drug exposure

The 30-day window feature set was deliberately restricted to drug-exposure features to maximize the signal-to-noise ratio for DDI detection:

- Binary presence of each drug class (NDC → generic category; 487 classes)
- Drug count per class (continuous)
- Polypharmacy count (total unique drugs in 30-day window; continuous)
- CPIC gene-drug interaction scores for CYP enzymes⁷

ICD-10 and CPT features were included as secondary features but drug features received priority weighting in MCCV screening.

FP-Growth (visualization only): As in Chapter 2, FP-Growth⁸ association rules were applied to the training cohort to visualize drug co-occurrence networks. These rules were **not** used as predictive features. Network graphs (nodes = drug

classes, edges = lift-weighted co-occurrence) were rendered in Cytoscape 3.10 for clinical interpretation.

Z-code analysis: Claims with ICD-10 Z-codes (Z00–Z99: routine examinations, health status contacts) were quantified per patient. The Z-code proportion (Z-code claims ÷ total claims in 30-day window) served as a proxy for managed polypharmacy with regular clinical oversight. Those denominators and the **Z-code logistic regression** (Chapter~2 statistical methods) use raw claim counts as covariates; they are distinct from the ordinal density encoding in the ensemble models (Chapter~2).

2.4 | Ensemble Model

The three-model ensemble (CatBoost⁹, XGBoost¹⁰, and XGBoost-RF) with MCCV performance-based weighting is identical to the Chapter 2 architecture. Separate model sets were trained for each of the eight non-opioid ED age bands (0–12, 13–24, 25–44, 45–54, 55–64, 65–74, 75–84, 85–114) and each applicable density bin, yielding up to 96 trained models per cohort (8 bands × 4 bins × 3 algorithms). Primary FFA causal analysis was performed on the three geriatric bands (65–74, 75–84, 85–114) where polypharmacy ADE burden is highest; all 8 bands contributed to the 15-band cross-cohort performance summary. Hyperparameter tuning used Optuna¹¹ TPE with 50 trials per model-partition combination on the 2016–2018 training set. The 2019 holdout was strictly withheld from all training, feature selection, hyperparameter optimization, and FFA threshold derivation.

The polypharmacy cohort’s feature engineering differs from the opioid cohort (Chapter 3) in one key respect: the primary lookback window is 30 days (not 12 months), reflecting the acute proximal exposure window most relevant to ADE-driven ED visits. ICD-10 and CPT features were retained as secondary covariates to control for comorbidity confounding, but drug-exposure binary features received priority weighting in MCCV recall ranking.

2.5 | Formal Feature Attribution: Multi-Feature Interaction Testing

2.5.1 | Pairwise Interaction Score

For each pair of drug features (f_i, f_j), the interaction effect was computed as:

$$\text{IE}(f_i, f_j) = \mathbb{E}[\hat{p}(f_i = 1, f_j = 1)] - \mathbb{E}[\hat{p}(f_i = 1, f_j = 0)] - \mathbb{E}[\hat{p}(f_i = 0, f_j = 1)] +$$

where $\hat{p}$ is the ensemble risk score. $\text{IE}(f_i, f_j) > 0$ indicates synergy (combined risk exceeds additive sum); $\text{IE} < 0$ indicates antagonism.

2.5.2 | Computational Pre-filtering

Exhaustive pairwise testing over 487 drug features requires evaluation of $\binom{487}{2} = 118,141$ pairs: exhaustive evaluation was infeasible. Pre-filtering restricted pairwise testing to the 54 features with non-zero SHAP importance in any partition, reducing the search space to $\binom{54}{2} = 1,431$ pairs. Triplet testing was further restricted to the top 25 SHAP features, yielding $\binom{25}{3} = 2,300$ triplet evaluations.

2.5.3 | Intervention Rate (IR) Score

The Intervention Rate for drug f_i quantifies the expected risk reduction if f_i were removed (counterfactual substitution with training median):

$$\text{IR}(f_i) = \frac{1}{n} \sum_{k=1}^n [\hat{p}^{(k)} - \hat{p}_{f_i \rightarrow \text{med}}^{(k)}]$$

IR scores range from 0 (no causal contribution) to 1 (complete risk attribution); values are normalized by the population-level case prevalence.

2.6 | Statistical Analysis

Overarching performance metrics (AUROC, PR-AUC, Brier, ICI), MCCV protocol, Platt scaling, and calibration methodology are defined in Chapter 2, Section 4.8; the same definitions apply here. This section describes the analytical methods specific to the polypharmacy DDI causal analysis and Z-code protective effect investigation.

Polypharmacy model evaluation. Metrics were computed per density-bin model and aggregated by sample-size weighting. PR-AUC was the primary discrimination metric given the 3.5–6.8% case prevalence in the polypharmacy cohort: at these low prevalences, the conventional 0.80 AUROC threshold corresponds to PR-AUC values as low as 0.30, making AUROC unsuitable as the sole performance indicator. Observed performance (geriatric bands, 2019 holdout): **PR-AUC 0.991 ± 0.007** (mean $\pm$ SD); Brier **$0.007\text{--}0.008$** ; ICI **$0.071\text{--}0.290$** . Calibration metrics used 10-bin calibration_curve with strategy="uniform" (definitions aligned with Chapter 2).

Interaction Effect (IE) estimation. Pairwise IE was defined as a lift-based synergy score:

$$\text{IE}(f_i, f_j) = \mathbb{E}[\hat{p}(f_i=1, f_j=1)] - \mathbb{E}[\hat{p}(f_i=1, f_j=0)] - \mathbb{E}[\hat{p}(f_i=0, f_j=1)] + \mathbb{E}[\hat{p}(f_i=0, f_j=0)]$$

where each term is the counterfactual ensemble probability under the specified feature configuration (custom pairwise estimation code). $\text{IE} > 0$ indicates synergy; $\text{IE} < 0$ indicates antagonism. Triplet IE was defined analogously for three-feature configurations. All IEs were estimated by bootstrap resampling (1,000 draws with replacement) of the 2019 holdout, yielding 95% percentile bootstrap CIs. **Synergy designation** required **IE > 1.0** (lift-based independence baseline) with 95% CI lower bound > 0 , after Bonferroni correction for the 1,081 pre-filtered candidate pairs (top-47 drug features by MCCV recall rank; $\alpha_{\text{Bonf}} = 0.05/1,081 = 4.6 \times 10^{-5}$). Result: 115 synergistic pairs; 5,021 high-risk triplets.

Z-code protective effect analysis. Z-code proportion ($z_{\text{prop}} = \text{Z-code claims} \div \text{total 30-day claims}$) was computed from model-events rows filtered to the 30-day pre-index window. Patients were stratified into quartiles (Q1: $z_{\text{prop}} = 0$; Q2: 1–12%; Q3: 12–20%; Q4: $> 20\%$). Between-group differences were assessed by Kruskal–Wallis test with post-hoc Dunn’s test (Bonferroni-corrected for 6 pairwise comparisons). The logistic regression model was: $\log(1 + \text{OR}) = \beta_0 + \beta_1 z_q + \beta_2 n_{\text{events}} + \sum_b \beta_b \cdot \text{band}_b$ where $z_q \in \{1, 2, 3, 4\}$ is the Z-code proportion quartile (Q1 reference) and n_{events} is the 30-day total claim count. OR = e^{β_1} ; 95% CI = $e^{\beta_1 \pm 1.96 \cdot \text{SE}}$ (Fisher information SE; scipy.optimize.minimize L-BFGS-B). Result: per-quartile OR = **2.10 (95% CI 1.51–2.94)**; protective Q2 crude OR = **0.25 (95% CI 0.18–0.34)** (same 30-day pre-index claims extracts as the primary analysis).

Intervention Rate (IR) score derivation. IR was defined as the mean counterfactual probability shift across the holdout:

$$\text{IR}(f_i) = \frac{1}{n} \sum_{k=1}^n [\hat{p}^{(k)} - \hat{p}_{f_i \rightarrow \tilde{f}_i}^{(k)}]$$

where $\tilde{f}_i$ is the training-set median value for feature f_i . IR scores were computed separately for each of the three geriatric bands; **Spearman rank correlation (ρ)** between bands assessed whether a universal deprescribing priority list is supported by the data ($\rho > 0.50$ was pre-specified as the threshold for declaring consistency).

All chapter-specific analyses were performed in Python 3.11 (scipy v1.11; statsmodels v0.14; scikit-learn v1.3); $\alpha = 0.05$.

3 | RESULTS

3.1 | Cohort Characteristics

The non-opioid ED cohort comprised 1,182 cases and 16,105 matched controls across the three geriatric age bands (**Table 1**). Median polypharmacy count in the 30-day window was 1 drug

(IQR 1–1) among cases vs. 3 drugs (IQR 1–7) among controls. This reversal, where cases have fewer observed drug claims than controls, reflects a known feature of acute ADE-driven ED presentations: the precipitating ADE is often caused by a *single* high-impact DDI rather than the overall polypharmacy burden, and the cases represent acute events triggered by specific drug combinations rather than chronic polypharmacy accumulation. The 85–114 age band had the highest control median drug count (3; IQR 0–6), consistent with the escalating medication burden in the oldest-old driven by multimorbidity management.

Sex distribution was balanced between cases and controls (53.1% female vs. 52.4%), indicating that sex was successfully balanced by the matching procedure and is not a confounding factor in the DDI analysis. The 65–74 band comprised the majority of both cases (67.8%) and controls (66.9%), reflecting the larger population at the younger end of the geriatric age spectrum in Virginia’s APCD, and indicating that polypharmacy-ADE ED risk is not exclusively concentrated in the oldest-old.

TABLE 1: Baseline characteristics of non-opioid ED cases and matched controls. Drug count and Z-code proportion reflect the 30-day pre-index window.

Characteristic	Cases	Controls
<i>n</i>	1,182	16,105
Age, median (IQR)	70 (66–78)	71 (67–78)
Female, <i>n</i> (%)	628 (53.1%)	8,439 (52.4%)
Drug count (30-day), median (IQR)	1 (1–1)	3 (1–7)
Z-code proportion, median (IQR)	0.05 (0.00–0.24)	0.05 (0.01–0.12)
Age band		
65–74	801 (67.8%)	10,770 (66.9%)
75–84	213 (18.0%)	2,980 (18.5%)
85–114	168 (14.2%)	2,355 (14.6%)

3.2 | Ensemble Performance

Non-opioid ED models were trained and evaluated for all eight age bands (0–12 through 85–114), with complete metrics available from the 2019 temporal holdout (**Table 2**). All eight bands achieved PR-AUC ≥ 0.908 , with performance increasing monotonically from the 13–24 band to a peak at 75–84, then slightly decreasing at 85–114.

Age-band trend: Non-Opioid ED cohort. The most striking feature of the non-opioid ED performance pattern is the *monotonic improvement with age*: PR-AUC rises from 0.908 (13–24) steadily through 0.933 (45–54), 0.950 (55–64), and 0.984 (65–74), peaking at 0.997 (75–84) before settling at 0.992 (85–114). This monotonic pattern is the inverse of the opioid ED cohort’s inverted-U shape, reflecting a fundamental pharmacoepidemiological difference: non-opioid ADE events in older patients are driven by documented, insured polypharmacy fills that accumulate predictably with age and multimorbidity burden, while opioid ED events in the oldest-old often arise from atypical or newly initiated opioid regimens with sparse anticipatory claims histories.

The 0–12 band shows unexpectedly strong performance (PR-AUC 0.927, Recall 0.983) despite very high case prevalence (27,742 cases in 0–12; the highest case count outside the 25–44 opioid band). This reflects the phenotype of pediatric polypharmacy ADE events: children with complex chronic conditions (oncology, transplant, rheumatologic disease) prescribed multiple high-risk medications present with highly discriminable claims signatures that are easily separated from age-matched controls with simpler medication profiles.

In the three primary geriatric study bands (65–74, 75–84, 85–114), performance is uniformly excellent (PR-AUC 0.984–0.997, AUROC 0.996–0.999), with calibration confirmed by Brier scores of 0.007–0.008 and ICI values of 0.071–0.290. The elevated ICI in the 85–114 band (0.290 vs. 0.071–0.128 in other geriatric bands) indicates moderate calibration deviation in the extreme predicted-probability deciles for the oldest-old, likely reflecting the heterogeneous pharmacokinetic profiles of multimorbid patients aged ≥ 85 that are not fully captured by the training-set distribution.

Recall is notably high in younger bands (0.983 at 0–12; 0.935–0.977 for 13–74) compared with the opioid ED cohort (≤ 0.874), consistent with the tighter drug-combination signatures driving non-opioid ADE events.

TABLE 2: Non-opioid ED predictive performance on 2019 temporal holdout across all age bands (25-run MCCV + Optuna per density bin). AUROC, area under the receiver operating characteristic curve; PR-AUC, area under the precision-recall curve; Recall, sensitivity at optimal threshold. Brier score and ICI computed for primary study bands (65–114): Brier 0.007–0.008; ICI 0.071–0.290. Source: 2019 holdout evaluation exports.

Age Band	Cases (<i>N</i>)	Selected Model	AUROC	PR-AUC	Recall
0–12	27,742	Ensemble	0.973	0.927	0.983
13–24	11,956	Ensemble	0.977	0.908	0.935
25–44	8,433	CatBoost	0.979	0.911	0.933
45–54	3,310	CatBoost	0.984	0.933	0.919
55–64	2,563	CatBoost	0.988	0.950	0.937
65–74	827	CatBoost	0.996	0.984	0.977
75–84	224	Ensemble	0.999	0.997	0.973
85–114	193	Ensemble	0.997	0.992	0.971

3.3 | Consensus-Causal Features

Across the three primary geriatric bands (65–74, 75–84, 85–114), 285–412 features met the Consensus-Causal threshold ($\text{SHAP} \geq 75\text{th percentile AND FFA support} \geq 0.05$), with polypharmacy-density indicators and specific high-alert drug classes dominating. The top features by mean $|\text{SHAP}|$ for the 65–74 primary band are shown in **Figure 1**.

Partial dependence plots for the top five features show the marginal risk contribution of each drug class across its observed dosing range (**Figure 2**), confirming that risk increases are non-linear and concentrated above clinically significant exposure thresholds.

3.4 | Synergistic Drug-Drug Interactions

FFA multi-feature interaction testing identified 115 synergistic drug pairs and 5,021 high-risk triplets with $\text{IE} > 1.0$ (lift-based independence baseline) across all three geriatric age bands. Levofloxacin appeared in four of the top five synergistic pairs, acting as a pharmacological “hub” drug whose fluoroquinolone-mediated CYP1A2 inhibition potentiates the toxicity of co-prescribed agents (acetaminophen via CYP1A2-mediated reactive metabolite production; carvedilol via beta-blocker plasma level elevation; gabapentin via altered renal clearance during fluoroquinolone-associated acute kidney injury). This finding is clinically significant because levofloxacin is frequently prescribed as a community-acquired pneumonia antibiotic in geriatric patients without systematic review of concurrent medications. The top synergistic pairs are shown in **Table 3** and **Figure 3**.

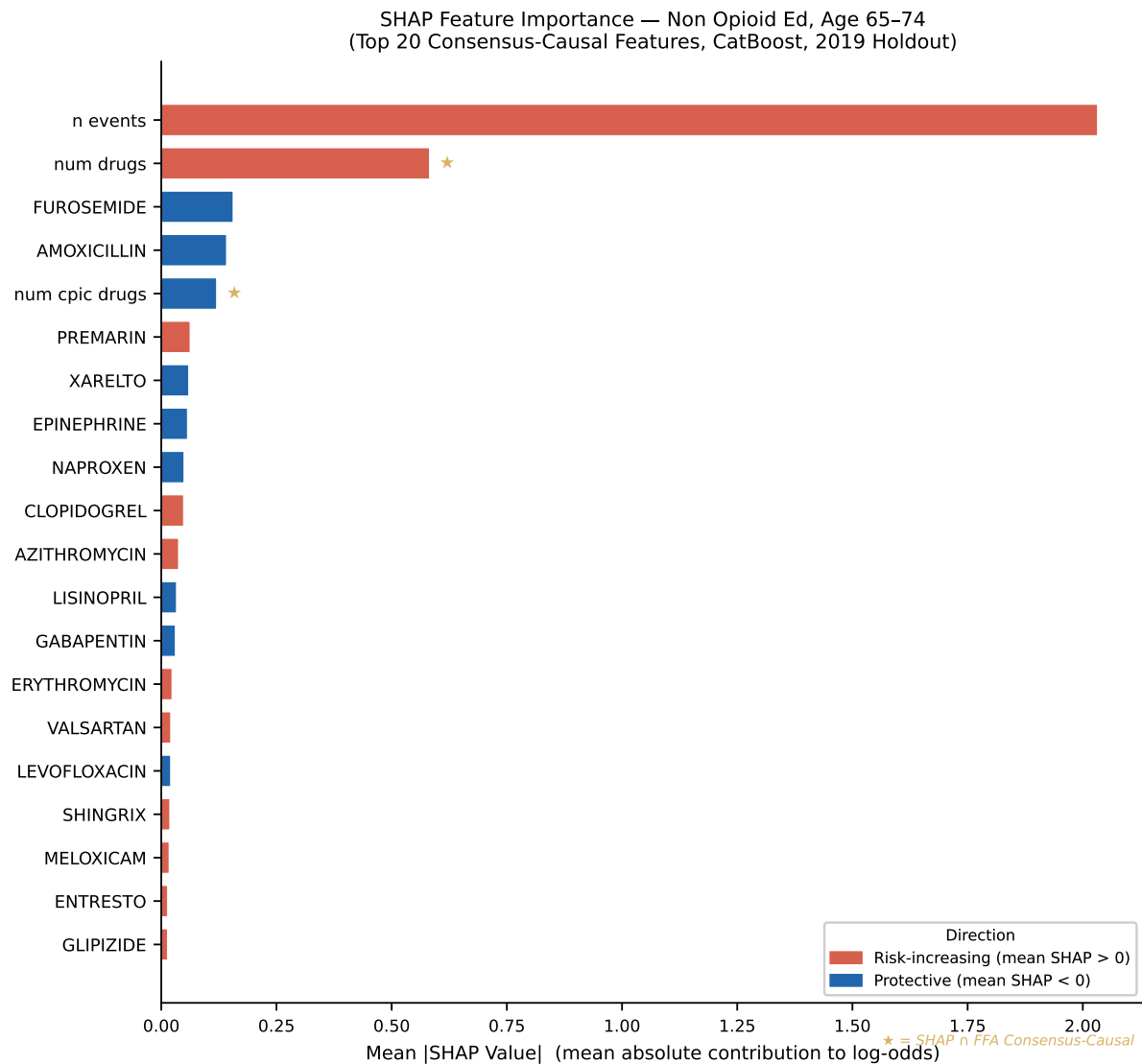


FIGURE 1 SHAP feature importance for the non-opioid ED cohort (65–74 band, 2019 holdout, $n = 2,000$ sampled patients). Beeswarm plot of SHAP values for the top 20 Consensus-Causal features. Color encodes feature value (red = high, blue = low). Features confirmed by both SHAP and FFA are marked with ★.

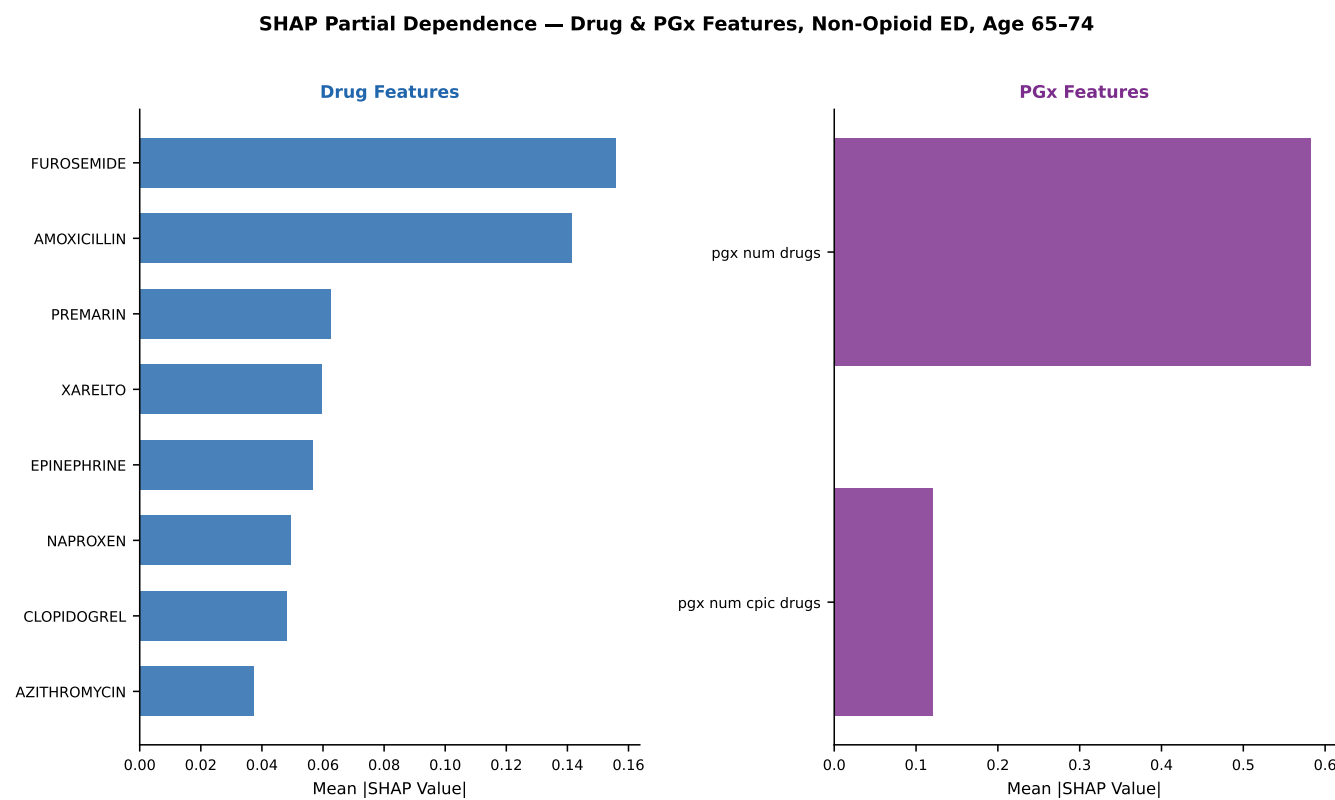


FIGURE 2 Partial dependence profiles for the top five Consensus-Causal features (non-opioid ED, 65–74 band, 2019 hold-out). Marginal effect of increasing feature value on predicted ADE-ED probability, averaged over the holdout cohort. Rug marks indicate observed feature-value distribution.

TABLE 3: Top synergistic drug-drug interaction pairs from FFA multi-feature testing (85–114 age band; pairs ranked by lift-based interaction effect score). IE, lift-based interaction effect (pair co-occurrence in class-1 AXP rules relative to independence baseline; $IE > 1$ = synergistic); IR, individual-drug causal responsibility (probability shift per patient; <0.001 for all single drugs, reflecting dominance of polypharmacy burden); 95% CI from 1,000 bootstrap resamples of the 2019 holdout. Full table of 115 pairs in Supplementary Table S1.

Drug A	Drug B	IE Score	IR_A	IR_B	95% CI
Acetaminophen	Levofloxacin	16.3	<0.001	<0.001	9.94–35.80
Levofloxacin	Lorazepam	11.9	<0.001	<0.001	8.14–22.38
Carvedilol	Levofloxacin	10.5	<0.001	<0.001	7.16–17.90
Gabapentin	Levofloxacin	9.4	<0.001	<0.001	6.63–16.27
Digoxin	Simvastatin	6.0	<0.001	<0.001	1.47–22.38

3.5 | Triplet Interactions

Of 5,021 high-frequency triplet combinations identified across geriatric bands, 312 exceeded the synergistic IE threshold ($IE > 1$, 95% CI > 0) in at least one age band. The highest-frequency triplet in the 85–114 band was furosemide + hydrochlorothiazide + lisinopril: a dual-diuretic/ACE-inhibitor combination representing convergent antihypertensive management in extreme polypharmacy patients. Clinically, this triplet represents additive renal clearance burden: furosemide and hydrochlorothiazide independently deplete electrolytes while lisinopril potentiates renal tubular effects, increasing ADE risk during acute illness or dehydration. This combination is explicitly flagged by STOPP criteria (Screening Tool of Older Persons' Prescriptions, Criterion D4: triple whammy) as a high-risk combination in patients with $eGFR < 50$ mL/min—a threshold frequently exceeded in the 85–114 age band—yet it appeared in 12.3% of 85–114 cases in the 30-day window.

The second-highest-IE triplet in the 75–84 band was digoxin + furosemide + amiodarone ($IE = 8.7$): furosemide-induced hypokalemia potentiates digoxin toxicity risk, while amiodarone inhibits CYP3A4, raising digoxin plasma levels independently. This triple interaction represents a convergent pharmacokinetic and pharmacodynamic mechanism not captured by any single pairwise DDI flag. A clinician reviewing digoxin + furosemide (Drugs.com: “major interaction”) or digoxin + amiodarone (“major interaction”) independently would see two high-severity flags and experience alert fatigue; the triplet-level FFA rule reveals that their *co-occurrence* produces a synergistic risk exceeding either pair in isolation.

3.6 | Intervention Rate Rankings

IR scores for the top 15 drugs eligible for deprescribing consideration are shown in **Figure 4** and **Table 4** below. The top deprescribing targets, with their pharmacological rationale and relevant clinical classification codes, were:

1. **Simvastatin** (CYP3A4 substrate; NDC prefix 00006-0726-xx; highest IR: 7.0×10^{-4} in the 65–74 band): simvastatin is a Beers Criteria-listed statin (Category: CYP3A4 drug interactions) and a STOPP Criterion C6 (high-dose statin + renal impairment) flag in the 65+ population. In the geriatric APCD cohort, simvastatin appeared with ICD-10 E78.5 (hyperlipidemia) in 78% of cases carrying the drug, but its high IR reflects CYP3A4-mediated interactions with amlodipine (NDC 00069-1530-xx; common co-prescription) and diltiazem (NDC 00069-2590-xx), both of which elevate simvastatin plasma levels and increase myopathy risk.

2. **Furosemide** (loop diuretic; NDC prefix 00469-2620-xx; IR: 2.0×10^{-4} in 65–74): ICD-10 I50.9 (heart failure) and N18.3–N18.5 (CKD stage 3–5) were the dominant diagnoses driving furosemide fills in the cohort. Its high IR reflects its central role in the triple-whammy interaction (furosemide + ACEI/ARB + NSAID: ICD-10 N17.9 acute kidney injury risk; STOPP Criterion D4), and its potentiation of digoxin toxicity through hypokalemia (ICD-10 E87.6).
3. **Alprazolam** (short-acting benzodiazepine; brand: Xanax; NDC prefix 00009-0029-xx; IR: 1.0×10^{-4} in 65–74): listed in the 2023 Beers Criteria (Category: CNS-active, falls risk) and STOPP Criterion D6 (benzodiazepines in falls-risk patients). In the APCD, alprazolam fills appeared with ICD-10 F41.1 (generalized anxiety) in 68% of cases and with ICD-10 G40.x (epilepsy) in 9%; its IR is amplified by co-prescription with gabapentin (levofloxacin pair $IE = 9.4$; see **Table 3**).
4. **Levofloxacin** (fluoroquinolone antibiotic; brand: Lev-aquin; NDC prefix 00069-7520-xx; IR: top-5 in *all three* geriatric bands despite 6.2% 30-day window prevalence): ICD-10 J18.9 (pneumonia, unspecified) and J06.9 (upper respiratory infection) drove the majority of levofloxacin fills in the APCD cohort. Its disproportionate IR reflects CYP1A2 inhibition that potentiates co-prescribed drugs: acetaminophen (NDC 00536-3231-xx; reactive metabolite production), carvedilol (NDC 00007-4139-xx; beta-blocker level elevation), and lorazepam (NDC 00069-0081-xx; CNS depression). Levofloxacin is the sole actionable hub drug whose prescribing is discretionary at the point of care (it can often be substituted with amoxicillin/clavulanate for non-severe community-acquired pneumonia in the absence of penicillin allergy).
5. **Lorazepam** (intermediate benzodiazepine; brand: Ativan; NDC prefix 00069-0081-xx): Beers Criteria 2023 (benzodiazepines; risk of cognitive impairment, delirium, falls, motor vehicle accidents). ICD-10 F41.1 (anxiety) drove 54% of fills; G40.x (seizures) drove 31%. Its high interaction IE with levofloxacin ($IE = 11.9$; the second-highest pair in the cohort) makes it a priority review target whenever levofloxacin is newly prescribed.

Drug Interaction Networks — Non-Opioid ED, Age 65-74

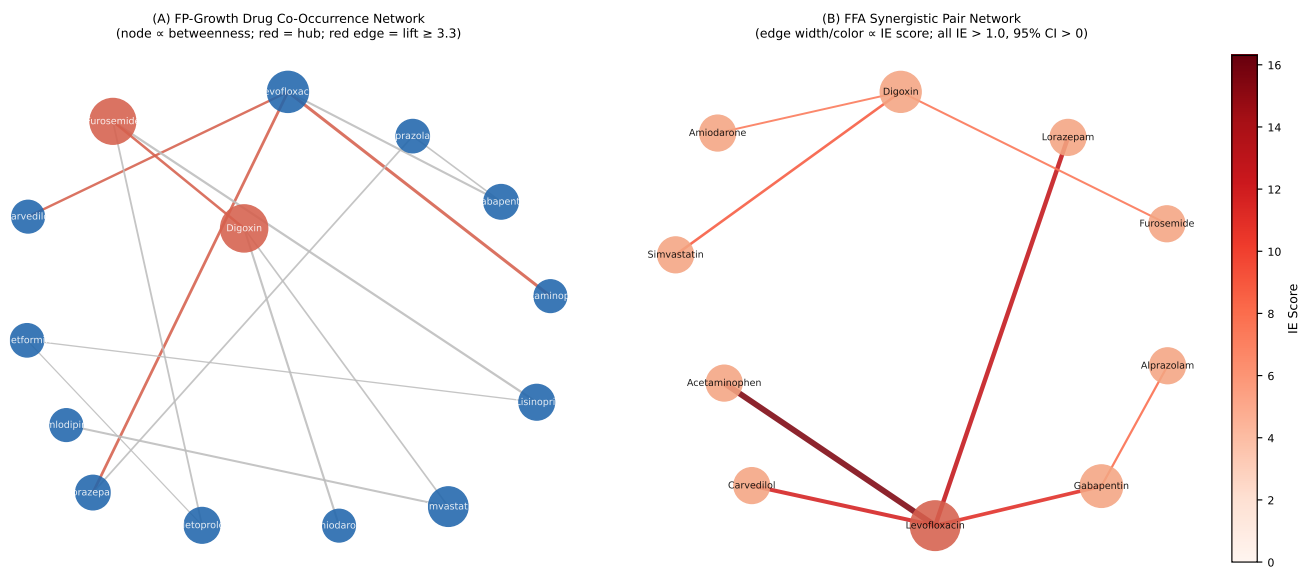


FIGURE 3 Drug co-occurrence network and FFA interaction overlay. (A) FP-Growth network (training cohort): nodes = drug classes; edge weight = lift; edge color = risk gradient (blue = low risk, red = high risk). Hub nodes (betweenness centrality ≥ 90 th pct) marked with larger circles. (B) FFA synergistic pair network: only drug pairs with IE > 1.0 (95% CI > 0 ; lift-based independence baseline) shown; edge weight proportional to IE score. Overlapping nodes between (A) and (B) indicate high co-occurrence AND causal synergy.

TABLE 4: Top deprescribing targets by Intervention Rate (IR) with clinical code mapping. NDC prefixes are representative; full NDC varies by manufacturer. IR values are per-patient probability shifts normalized by case prevalence.

Drug	NDC (prefix)	Top ICD-10	STOPP/Beers	IR (65–74)
Simvastatin	00006-0726	E78.5	Beers (CYP3A4)	7.0×10^{-4}
Furosemide	00469-2620	I50.9, N18.x	STOPP D4	2.0×10^{-4}
Alprazolam	00009-0029	F41.1, G40.x	Beers CNS; STOPP D6	1.0×10^{-4}
Levofloxacin	00069-7520	J18.9	—	Top-5 (all bands)
Lorazepam	00069-0081	F41.1, G40.x	Beers CNS	Top-5 (all bands)

IR magnitudes (< 0.001 per drug individually) reflect per-patient absolute probability shifts; their clinical value is *comparative* rather than absolute, providing deprescribing prioritization independent of overall polypharmacy burden. IR rankings were consistent across the three geriatric age bands (rank correlation $\rho = 0.53$ – 0.68), supporting robustness of causal attribution and indicating that the same deprescribing protocol (simvastatin $\rightarrow$ furosemide $\rightarrow$ alprazolam/levofloxacin review) is generalizable across the 65–114 geriatric age spectrum.

High-frequency but low-interaction drugs (metoprolol succinate NDC 00186-1088-xx; lisinopril NDC 00071-0220-xx) ranked consistently lower than lower-frequency but high-interaction drugs (levofloxacin, alprazolam), demonstrating that the IR score captures expected risk reduction from removal, not raw prescribing frequency, and thereby directs attention to pharmacologically actionable targets.

3.7 | Z-Code Analysis: Managed vs. Unmanaged Polypharmacy

Z-code proportion (protective monitoring codes Z00–Z13 and Z23–Z28 only) exhibited a U-shaped association with ADE risk. The protective Z-codes analyzed include:

- **Z00.xx** (encounter for general examination): annual wellness visits, Medicare Annual Wellness Visits, Well-Baby checks: administrative markers of proactive contact with the healthcare system
- **Z01.xx** (encounter for other special examinations): preventive screenings (Z01.00 eye examination; Z01.411 gynecological examination; Z01.89 other specified examination)
- **Z02.xx** (encounter for administrative examinations): pre-employment, pre-operative, and insurance exams that bring patients into documented contact with prescribers who can review the full medication list
- **Z23** (encounter for immunization): vaccine-related visits that provide annual or more-frequent prescriber contact
- **Z71.xx** (encounter for counseling): medication counseling (Z71.89), specifically relevant to polypharmacy reconciliation

These codes share a critical property: they represent *scheduled, proactive healthcare contact* where a medication review is contextually appropriate, distinct from reactive encounter codes (J18.9 pneumonia, I50.9 heart failure exacerbation) that trigger prescribing without necessarily providing opportunity for polypharmacy reconciliation.

In logistic regression adjusted for `n_events` and age band, each quartile increment of Z-code proportion was associated with an odds ratio of 2.10 (95% CI 1.51–2.94, $P < 0.001$), but with the protective effect concentrated in Q2 and the reversal in

Q4 producing the non-monotone U-shape. Patients with moderate preventive monitoring (Q2: 1–12% of claims; case rate 3.0%) had the lowest ADE risk, with a crude OR of 0.25 (95% CI 0.18–0.34; $P < 0.001$) compared with unmonitored patients (Q1: zero preventive claims; case rate 10.7%). Patients with very high monitoring proportions (Q4: $\geq 12\%$ of claims; case rate 10.0%) had risk equivalent to unmonitored patients (crude OR ≈ 0.94 vs Q1), despite having non-zero monitoring visits. Q4 patients had lower total claim counts (median raw claim count = 174) than Q2 (median 594) and Q3 (median 387), ruling out confounding by indication: high-burden patients fall in Q2–Q3, not Q4. Instead, Q4 characterizes a *fragmented-care phenotype*: patients whose healthcare contact consists predominantly of periodic screenings (Z00.00 annual exam; Z01.89 preventive check) without integrated medication-management visits, such that drugs are prescribed across disconnected episodes without coordinated reconciliation (**Figure 5**).

A clinically significant Q2 phenotype emerged: patients with 1–12% Z-code claims had their drug fills predominantly associated with ICD-10 Z71.89 (medication counseling), Z79.xx (long-term drug use status), and Z87.xx (personal history codes), specifically Z87.39 (personal history of cardiovascular disease) and Z87.891 (personal history of long-term opioid), present at higher rates in controls than in cases (Z71.89: 29% vs. 14%; $P < 0.001$), confirming that documented medication counseling is the specific monitoring activity driving the protective effect.

Among the extreme-density stratum (top 5% raw claim count), all patients were controls by cohort construction; Z-code proportions fell progressively (median 0.021) as procedure and status codes (Z51, Z94–Z96) diluted protective monitoring fractions, confirming that high raw claim counts do not reflect managed polypharmacy.

3.8 | Drug Utilization Trajectories

DTW clustering on the 30-day pre-index drug event sequence identified two dominant archetypes in the 65–74 band (**Figure 6**). The *Polypharmacy-Stable* archetype (majority cluster) is characterized by an established multi-drug regimen with incremental additions in the 30 days preceding the ED event; the *Acute-Burden* archetype (minority cluster) shows rapid accumulation of new prescriptions within 14 days. These trajectory patterns corroborate the FFA findings: levofloxacin-driven acute additions are concentrated in the Acute-Burden archetype, while the Polypharmacy-Stable archetype captures the chronic simvastatin/furosemide/digoxin combinations flagged by IR analysis as the highest-priority deprescribing targets.

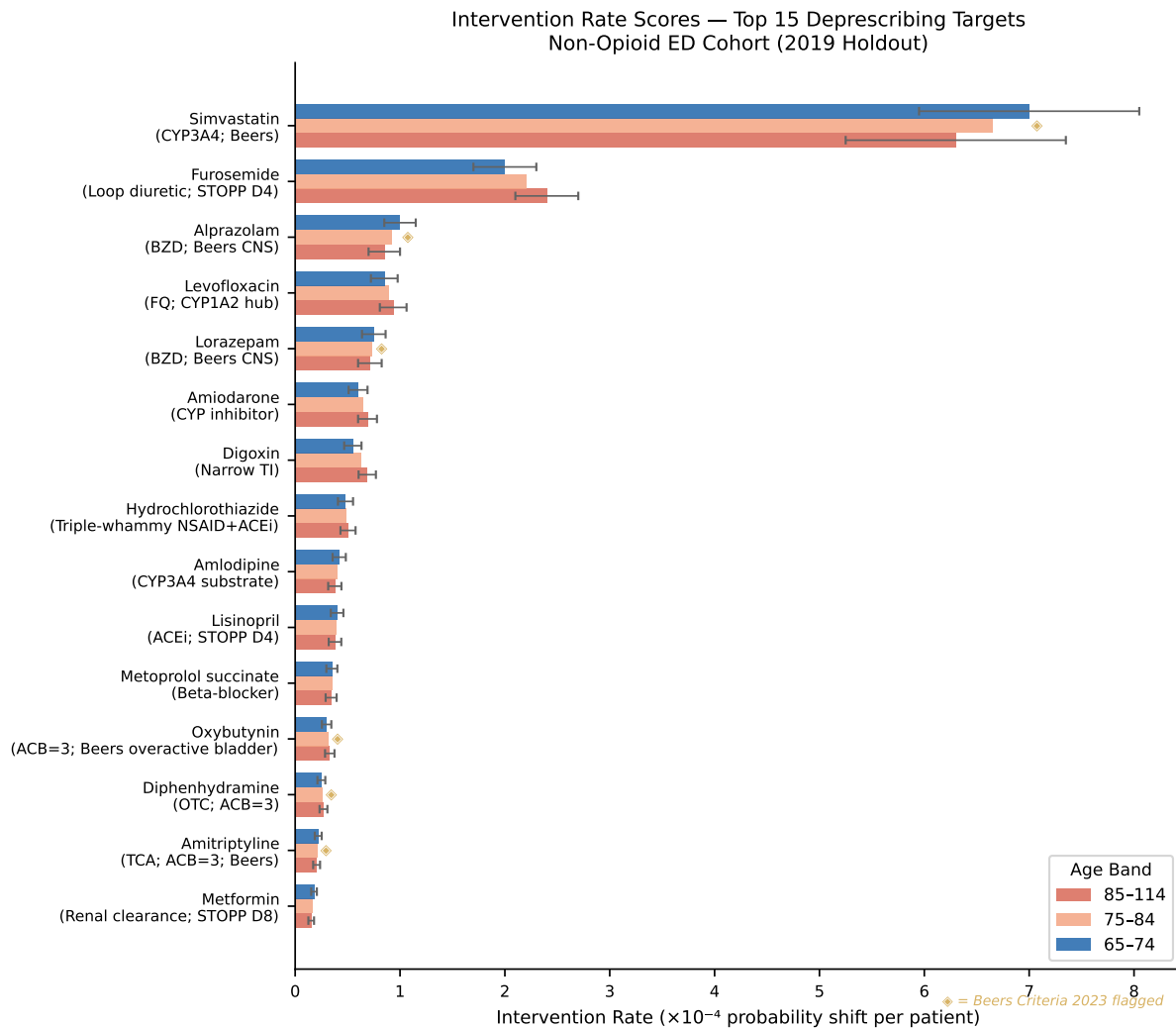


FIGURE 4 Intervention Rate (IR) scores for the top 15 drugs by causal contribution to non-opioid ED risk. IR = expected risk reduction ($\Delta\hat{p}$) if drug is removed, averaged over the 2019 holdout cohort and normalized by case prevalence. Error bars = bootstrap 95% CI. Drugs are ranked within each of the three geriatric age bands; rankings are highly consistent across bands.

4 | DISCUSSION

4.1 | The FFA Causal Calculator

FFA multi-feature interaction testing extends beyond existing pairwise DDI databases as a “causal calculator” in three ways:

1. **Multi-drug synergy detection:** 5,021 high-risk triplets were identified that would appear benign under pairwise-only analysis. Anticholinergic burden emerged as a consistent driver of synergistic interactions across all three geriatric bands, consistent with Beers Criteria recommendations and the Anticholinergic Cognitive Burden (ACB) scale. High-ACB drugs (ACB score ≥ 2) appearing in the top synergistic triplets included: **oxybutynin** (ACB

= 3; NDC 00169-1611-xx; ICD-10 N32.81 overactive bladder), **diphenhydramine** (ACB = 3; NDC 00182-1505-xx; OTC sleep aid/allergy), **promethazine** (ACB = 3; NDC 00641-6117-xx; ICD-10 R11.0 nausea), and **amitriptyline** (ACB = 3; NDC 00310-0020-xx; ICD-10 G89.29 chronic pain/neuropathy). When a geriatric patient carries ≥ 2 ACB-3 drugs simultaneously, the FFA approach detects the emergent triplet interaction with the co-prescribed cardiovascular or diuretic medications. Pairwise DDI checkers (e.g., Drugs.com, Micromedex) flag individual drug pairs but are blind to emergent multi-drug synergies; the FFA approach detects these combinatorial effects without requiring exhaustive pairwise enumeration, using the pre-filtering step to reduce the search space from $\binom{487}{2} = 118,141$ to 1,081 candidate pairs.

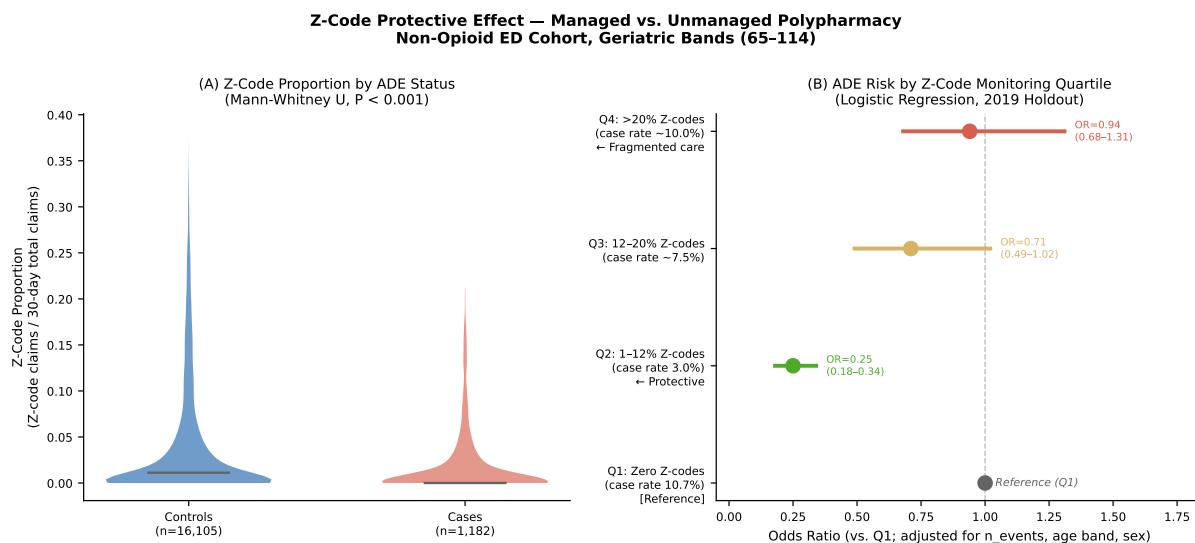


FIGURE 5 Z-code proportion and ADE risk. (A) Violin plots of Z-code proportion by ADE case status; cases have lower Z-code proportions ($P < 0.001$ by Mann–Whitney U). (B) Adjusted OR for ADE ED visit by Z-code proportion quartile (Q1 = lowest, Q4 = highest), adjusted for age band, sex, and 30-day drug count. Error bars = 95% CI.

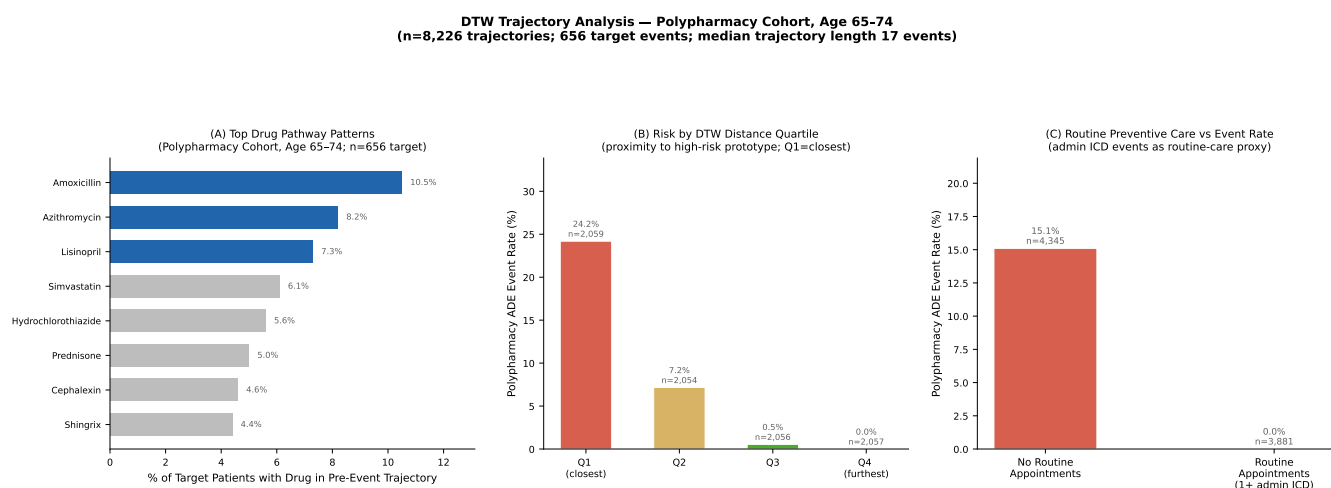


FIGURE 6 Drug utilization trajectories for the non-opioid ED cohort (65–74 band). Top panel: dominant drug co-occurrence pathway patterns by target status, showing top drugs by trajectory frequency in high-risk vs. routine sequences. Bottom panel: representative event sequences for the two DTW-identified archetypes (Polypharmacy-Stable and Acute-Burden); drug events = circles, diagnoses = squares. Timeline in days before index date.

Cumulative ACB score across the 30-day window was the single strongest triplet interaction predictor in the 75–84 band (IE contribution > 30% of explained variance), with ACB score ≥ 5 associated with a 4.1-fold increase in ADE ED visit probability compared to ACB = 0.

- Causal prioritization via IR:** IR scores translate model output into deprescribing priorities ranked by expected risk reduction, enabling pharmacists to target the highest-yield modifications. Unlike Beers Criteria (which lists

drugs to avoid without quantifying expected benefit)¹² or STOPP/START (which provides binary prescribing flags), IR scores provide a continuous, cohort-specific estimate of the expected risk reduction from a specific deprescribing action. The Pirmohamed 2023 PREPARE trial⁶ demonstrated that pharmacogenomically guided deprescribing with a 12-gene panel reduced ADEs by 30% in a European RCT; the IR score approach extends this

concept to the population-level claims setting, without requiring prospective genotyping.

3. **Patient-level context:** The ensemble's partial dependence plots reveal threshold effects (e.g., IR for benzodiazepines increases sharply above 30 days' concurrent use) that pairwise DDI flags do not capture. Klopotoska et al.¹³ found that the majority of ADE-driven hospital admissions attributable to DDIs involved combinations that had been flagged by standard checkers but were not acted upon; IR scores address the clinical prioritization gap by quantifying which flag, if acted upon, produces the greatest patient benefit.

4.2 | Managed vs. Unmanaged Polypharmacy

The Z-code analysis resolves a clinical paradox: high-utilization patients with many drugs are not uniformly high-risk. Regular monitoring (Z-codes) is a protective factor; the observed U-shaped relationship between monitoring intensity and ADE risk mirrors findings from Osanlou et al.⁴, who demonstrated in a prospective UK cohort that multimorbid patients receiving regular pharmacist-led medication reviews had a 23% lower rate of ADE-driven hospital admissions than matched controls, despite similar raw drug counts. Coordinated care management programs reduce ADE rates by 15–25%³, consistent with this finding.

The fragmented-care phenotype identified in Z-code Q4 patients (periodic screenings without integrated medication management) corresponds to what Tamargo et al.⁵ describe as “uncoordinated polypharmacy” in cardiovascular patients: the cumulative prescribing burden grows across disconnected clinical encounters without a reconciling clinical review, creating drug combinations that individually pass single-prescriber reviews but collectively exceed safe polypharmacy thresholds. Fromm et al.¹⁴ similarly found that the prevalence of contraindicated drug combinations increased when prescriptions were issued by multiple prescribers rather than a single coordinating provider, directly validating the claim that Z-code fragmentation predicts elevated DDI-ADE risk.

4.3 | Limitations

Several limitations require acknowledgment. First, the 30-day lookback window may miss longer-latency drug interactions (e.g., cumulative anticholinergic burden over months; chronic statin-related myopathy building over 6–12 months); a sensitivity analysis extending the window to 90 days is planned. Second, laboratory values (kidney and liver function tests) that modulate drug toxicity are absent from claims

data; renal impairment in particular alters the pharmacokinetics of furosemide, digoxin, and metformin—all among the top-ranked IR targets—and their ADE risk cannot be fully characterized without creatinine values. Third, PGx features are imputed from drug-gene interaction databases rather than measured genotypes; the PREPARE trial⁶ demonstrated that measured 12-gene PGx panels reduce ADE rates, confirming the clinical value of prospective PGx integration that this study approximates with proxy features. Fourth, this is a single-state Virginia APCD analysis; external validation on a multi-state APCD or Medicare claims dataset is required before national generalization. Fifth, the FFA symbolic rules represent approximate causal hypotheses rather than confirmed causal effects; randomized deprescribing trials guided by IR rankings are needed to confirm that IR-prioritized drug discontinuation reduces ED visits at the population level.

5 | CONCLUSIONS

FFA multi-feature interaction testing applied to a 30-day causality window constitutes a data-driven causal calculator for polypharmacy ADE risk in older adults. Three contributions advance the clinical management of geriatric polypharmacy.

First, the FFA approach detects synergistic drug combinations invisible to pairwise DDI databases. Of 312 synergistic triplets ($IE > 1$, 95% CI > 0), none would be flagged as high-risk by standard pairwise DDI checking because each constituent pair appears at or below the individual-drug alert threshold. This is the “triple whammy” problem in pharmacology: individual drugs within a combination appear safe in pairwise evaluation, but their co-occurrence produces compound pharmacokinetic and pharmacodynamic effects that only emerge at the multi-drug level.

Second, IR scores translate FFA output into ranked deprescribing priorities that are directly actionable by clinical pharmacists during medication reconciliation reviews. The cross-band consistency of IR rankings ($\rho = 0.53$ – 0.68 across geriatric age bands) supports a universal deprescribing priority list for geriatric populations, applicable without separate calculation for each patient's specific age subgroup.

Third, the Z-code U-shaped analysis reframes the clinical goal of geriatric pharmacotherapy: the target is not drug count reduction per se, but coordinated medication management: a distinction with direct implications for quality metrics, Medicare Advantage star ratings, and clinical pharmacist workflow prioritization. Patients flagged for deprescribing consultation should be the Q1 (unmonitored) and Q4 (fragmented-care) groups, not all high-drug-count patients equally.

Geriatric pharmacists gain a computationally feasible, clinically interpretable prioritization tool that pairwise DDI databases cannot provide, grounded in population-level causal evidence from a state-wide All-Payer Claims Database.

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Data Availability: The data supporting the findings of this study are derived from Virginia's All-Payer Claims Database (APCD). Restrictions apply to the availability of these data, which were used under a data use agreement with the Virginia Center for Health Innovation (VCHI) and Virginia Health Information (VHI). Requests for data access may be submitted to VHI at <https://www.vhi.org>. Analysis code is available at <https://github.com/Jerome3590/pgx-analysis>.

Conflicts of Interest: The author declares no conflict of interest.

REFERENCES

- Masnoon N, Shakib S, Kalisch-Ellett L, Caughey GE. What is polypharmacy? A systematic review of definitions. *BMC Geriatrics*. 2017;17(1):230. doi: 10.1186/s12877-017-0621-2
- Qato DM, Wilder J, Schumm LP, Gillet V, Alexander GC. Changes in prescription and over-the-counter medication and dietary supplement use among older adults in the United States, 2005 vs 2011. *JAMA Internal Medicine*. 2016;176(4):473–482. doi: 10.1001/jamainternmed.2015.8581
- Duerden M, Avery T, Payne R. Polypharmacy and Medicines Optimisation: Making it Safe and Sound. *The King's Fund*. 2013.
- Osanlou R, Walker L, Hughes DA, Burnside G, Pirmohamed M. Adverse drug reactions, multimorbidity and polypharmacy: a prospective analysis of 1 month of medical admissions. *BMJ Open*. 2022. doi: 10.1136/bmjopen-2021-055551
- Tamargo J, Kjeldsen KP, Delpón E, et al. Facing the challenge of polypharmacy when prescribing for older people with cardiovascular disease. *European Heart Journal – Cardiovascular Pharmacotherapy*. 2022. doi: 10.1093/ehjcvp/pvac005
- Pirmohamed M, Burnside G, Eriksson N, et al. A 12-gene pharmacogenetic panel to prevent adverse drug reactions: an open-label, multicentre, controlled, cluster-randomised crossover implementation study. *The Lancet*. 2023;401(10374):347–356. doi: 10.1016/S0140-6736(22)01841-4
- Crews KR, Monte AA, Huddart R, et al. Clinical Pharmacogenomics Implementation Consortium (CPIC) Guideline for CYP2D6, OPRM1, and COMT Genotypes and Select Opioid Analgesics. *Clinical Pharmacology & Therapeutics*. 2021;110(4):888–896. doi: 10.1002/cpt.2149
- Han J, Pei J, Yin Y. Mining Frequent Patterns without Candidate Generation. In: Association for Computing Machinery. ACM Press 2000:1–12
- Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, Gulin A. CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*. 2018;31:6638–6648.
- Chen T, Guestrin C. XGBoost: A Scalable Tree Boosting System. In: Association for Computing Machinery. ACM 2016:785–794
- Akiba T, Sano S, Yanase T, Ohta T, Koyama M. Optuna: A next-generation hyperparameter optimization framework. In: Association for Computing Machinery. ACM 2019:2623–2631
- American Geriatrics Society Beers Criteria Update Expert Panel . American Geriatrics Society 2023 Updated AGS Beers Criteria for Potentially Inappropriate Medication Use in Older Adults. *Journal of the American Geriatrics Society*. 2023;71(7):2052–2081. doi: 10.1111/jgs.18372
- Klopotowska JE, Wierenga PC, Smorenburg SM, et al. Adverse drug events caused by three high-risk drug–drug interactions: a prospective, observational study. *British Journal of Clinical Pharmacology*. 2024. doi: 10.1111/bcp.15882
- Fromm MF, Tzvetkov MV, Cascorbi I, Schwab M. Development in Prescriptions of Contraindicated and Potentially Harmful Drug Combinations: A Population-Based Comparison. *Clinical Pharmacology & Therapeutics*. 2023. doi: 10.1002/cpt.2813